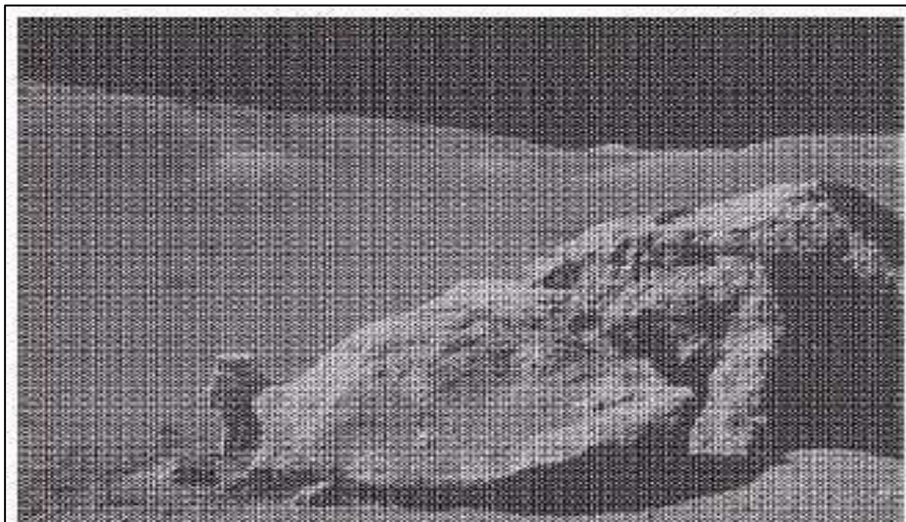
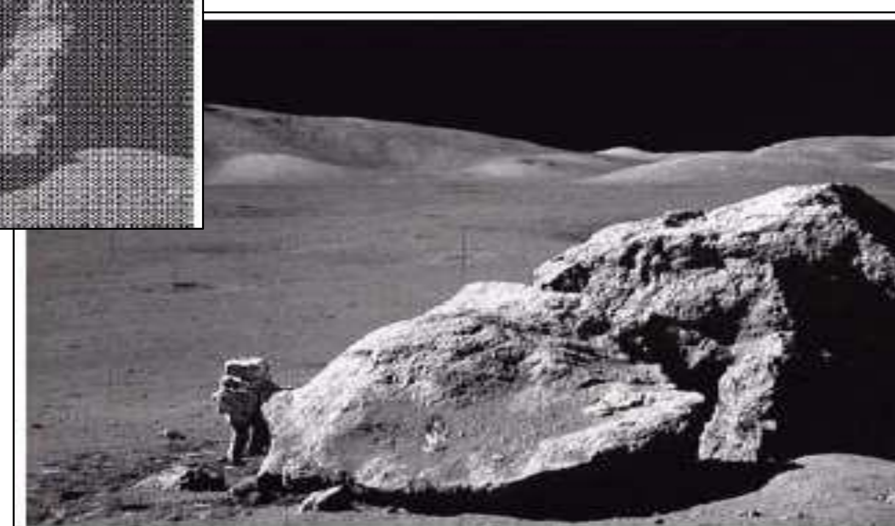
5 Image Restoration

- Improve an image in some predefined sense





Noise Models

- Source of Noise
  - Image acquisition (digitization)
  - Image transmission
- Gaussian Noise
  - Electronic circuit noise and sensor noise
- Impulse (Slat-and-Pepper) Noise
  - Quick transients, such as faulty switching during imaging
  - In practical, impulses are usually stronger than image signals

Spatial Filters for De-noising Additive Noise

Restoration in the Presence of Noise

1. Mean Filters

Arithmetic Mean Filter
$$\hat{f}(x,y) = \frac{1}{mn} \sum_{(s,t)\epsilon S_{xy}} g(s,t)$$

Geometric Mean Filter
$$\hat{f}(x,y)=\left[\prod_{(s,t)\epsilon S_{xy}} g(s,t)\right]^{\frac{1}{mn}}$$

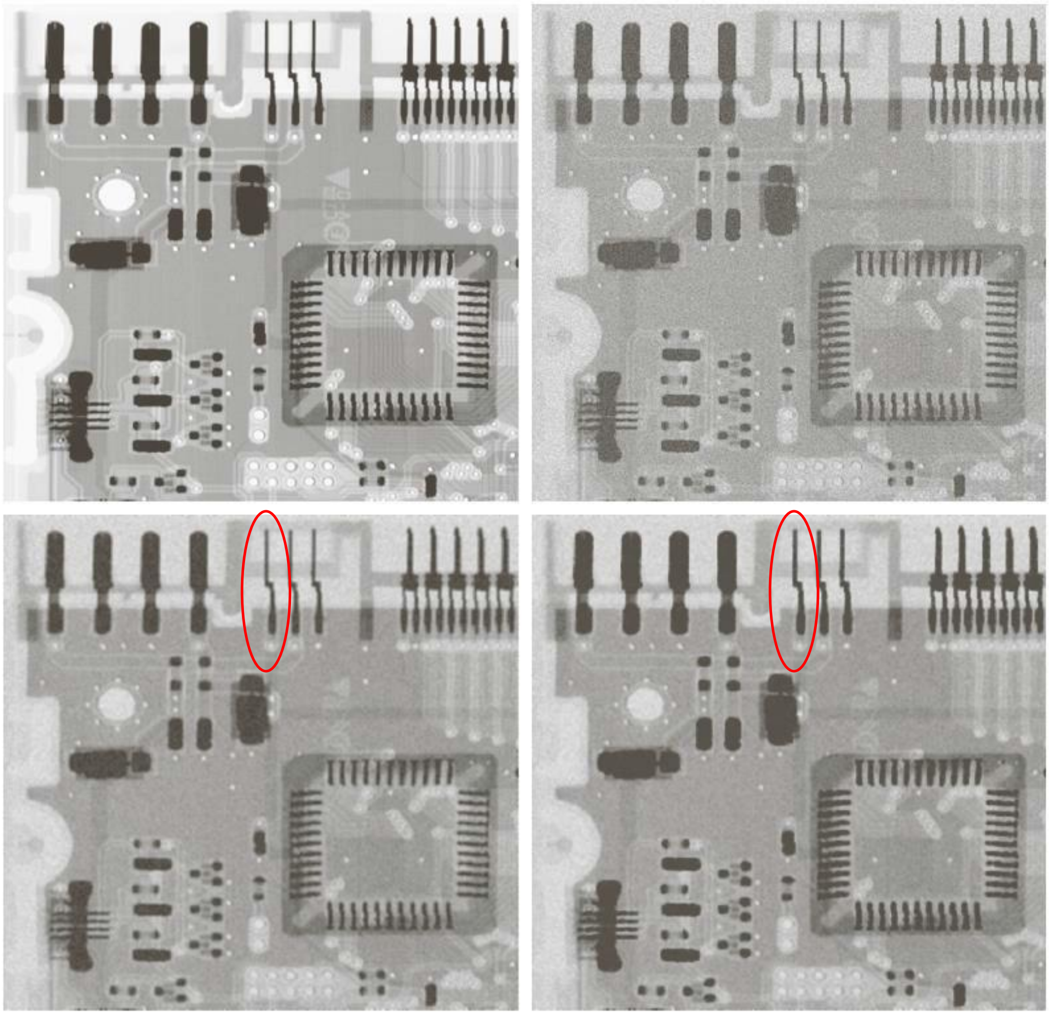
Harmonic Mean Filter
$$\hat{f}(x,y) = \frac{mn}{\sum_{(s,t)\epsilon S_{xy}} \frac{1}{g(s,t}}$$

Contra-harmonic Mean Filter
$$\hat{f}(x,y) = \frac{\sum_{(s,t)\epsilon S_{xy}} g(s,t)^{Q+1}}{\sum_{(s,t)\epsilon S_{xy}} g(s,t)^{Q}}$$

Mean Filters

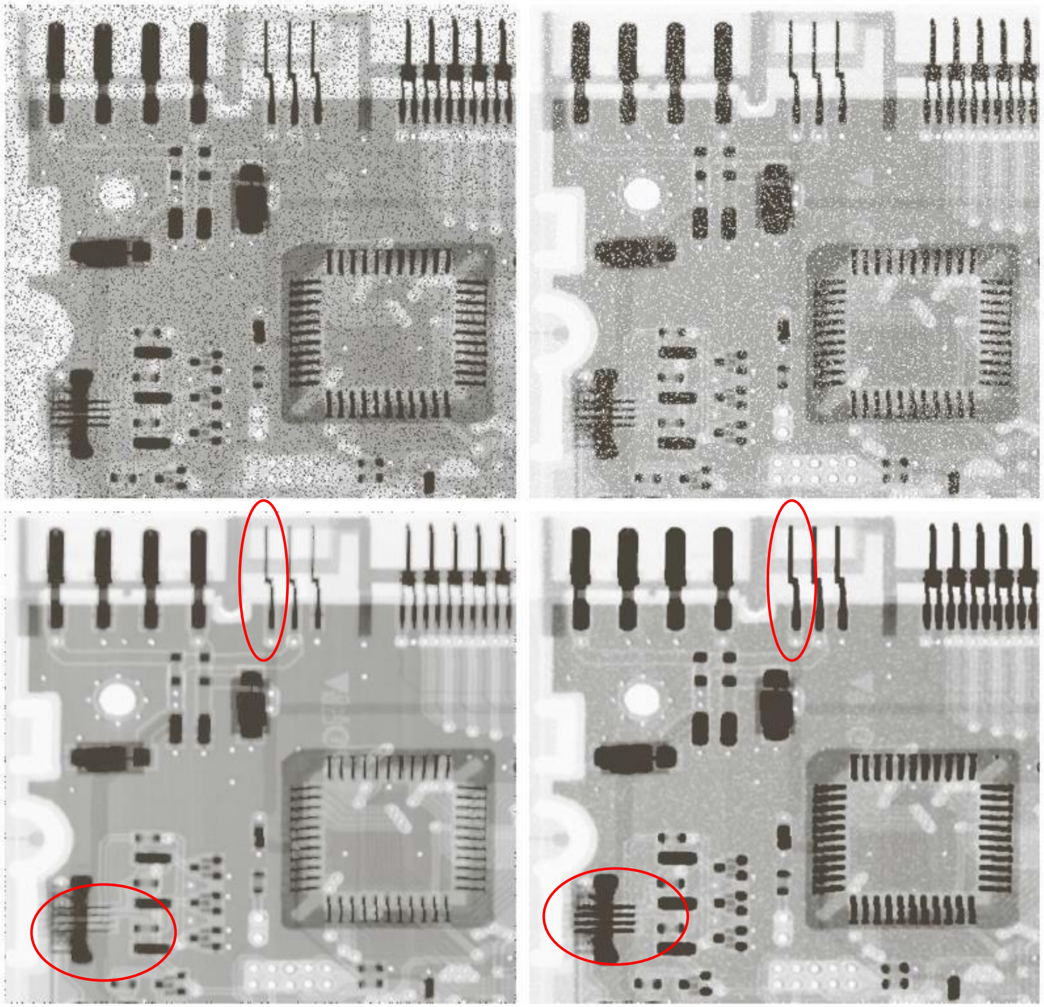
a. Original Image
b. Image corrupted by additive Gaussian Noise

c. After applying 3x3 Arithmetic mean filter.
d. After applying 3x3 Geometric mean filter.



Mean Filters

a. Image corrupted by additive pepper Noise
b. Image corrupted by additive salt Noise

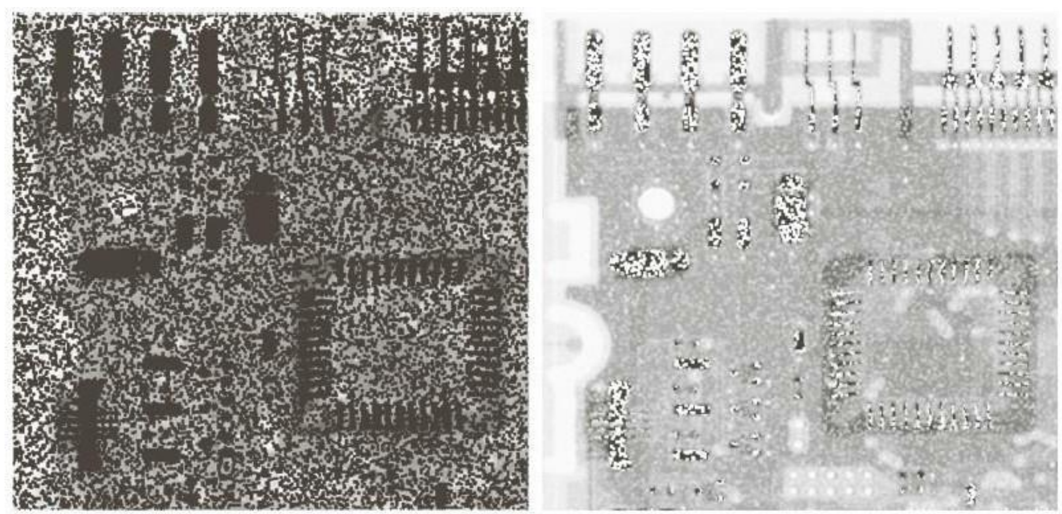


c. After applying 3x3 contra-harmonic mean filter with Q = 1.5 on a.
d. After applying 3x3 contra-harmonic mean filter with Q = -1.5 on b.

Mean filters

- Selecting the right value (and sign) of contra-harmonic mean filter is highly important.

Results of applying 3x3 contra-harmonic mean filter with Q = -1.5 on a. and Q = 1.5 on b.



Order-Statistics Filters

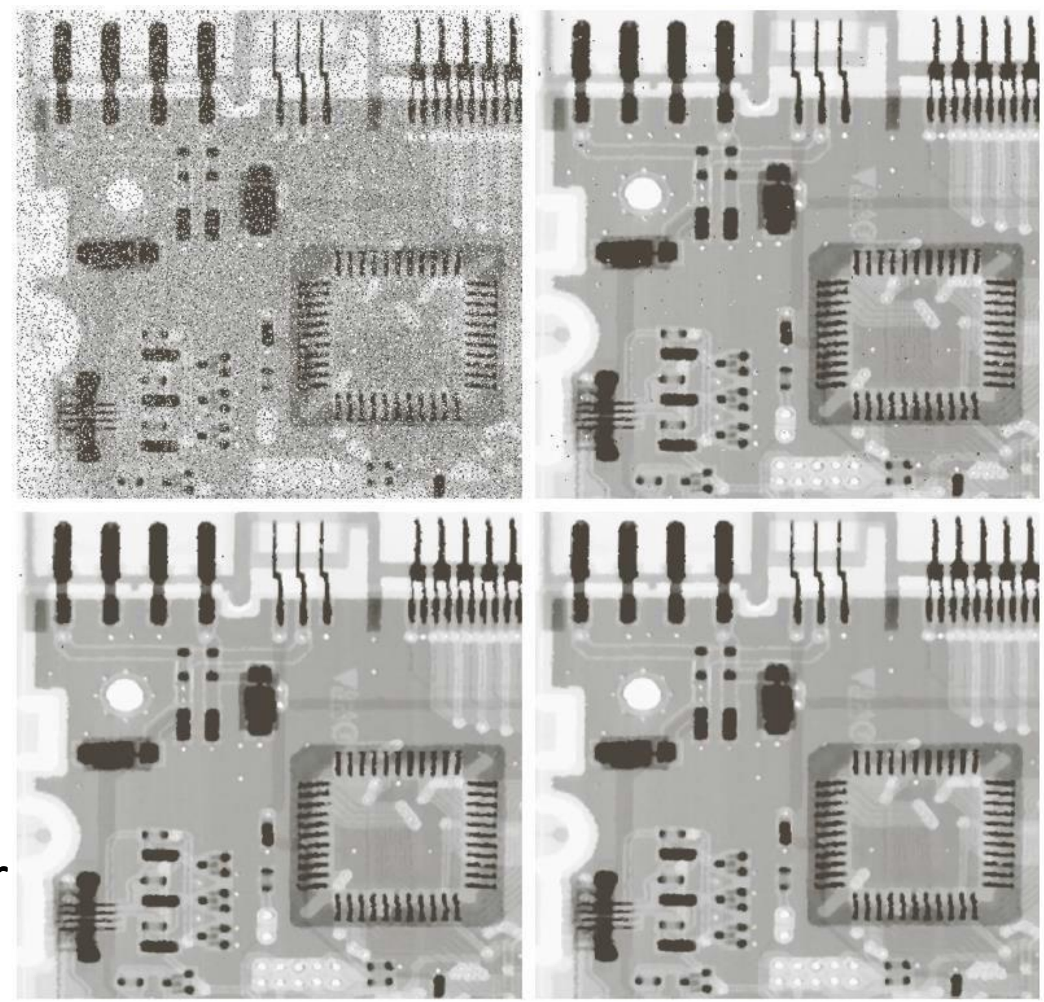
1. Median Filter $\hat{f}(x,y) = \mathrm{median}(\mathrm{g}(\mathrm{s},\mathrm{t}))$
2. Max Filter $\hat{f}(x,y) = \max(\mathrm{g}(\mathrm{s},\mathrm{t}))$
3. Min Filter $\hat{f}(x,y) = \min(\mathrm{g}(\mathrm{s},\mathrm{t}))$
4. Alpha-trimmed Mean Filter $$\hat{f}(x,y) = \frac{1}{mn-d} \sum_{(s,t)\epsilon S_{xy}} g_r\,(s,t)$$

   We delete the *d*/2 lowest and the *d*/2 highest gray-level values of $g(s, t)$. $g_r(s, t)$ represent the remaining *mn*-*d*.

Order-Statistics Filters

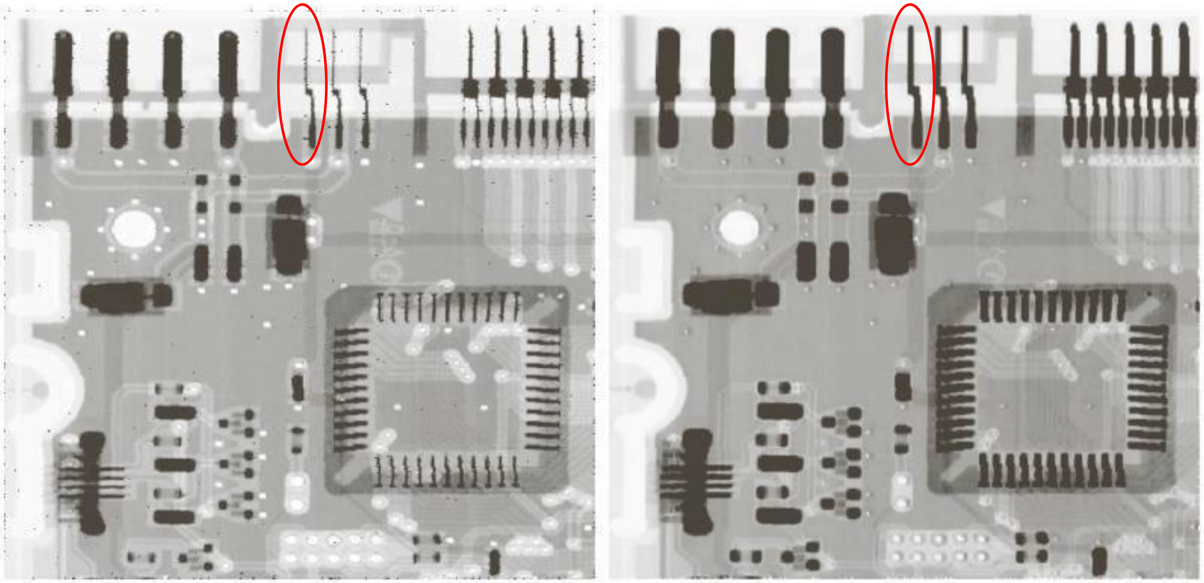
a. Image Corrupted with slat and pepper noise
b. After applying 3x3 median filter

c. After applying the same 3x3 median filter on b.
d. After applying the same 3x3 median filter on c.



Order-Statistics Filters

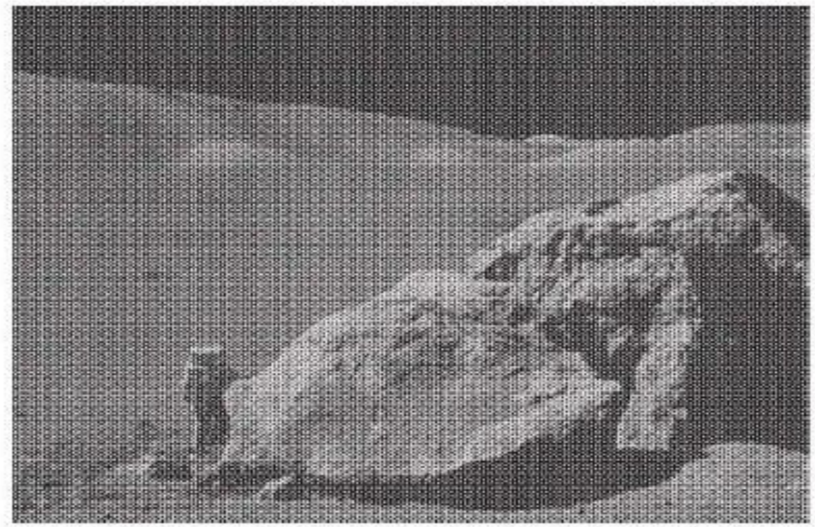
a. After applying Max filter.
b. After applying Min filter.

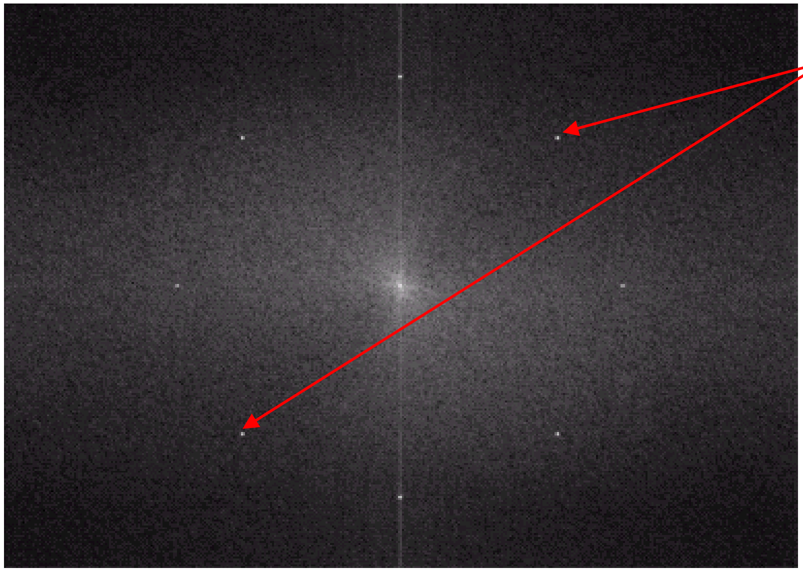


Periodic Noise

- Periodic noise arises from electrical or electromechanical interference during image acquisition
- It can be observed in the frequency domain

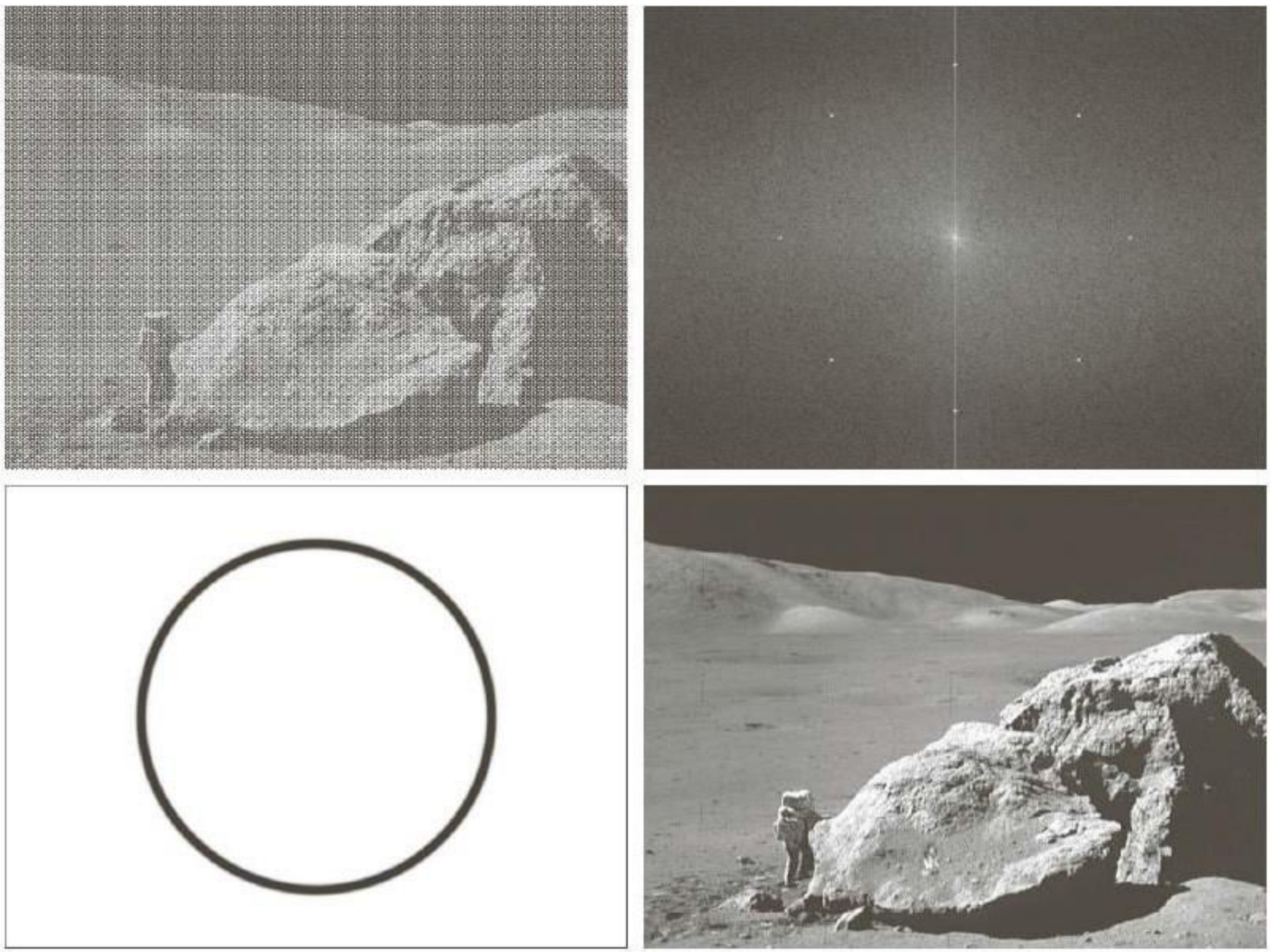
Periodic Noise Reduction





Sinusoidal noise:
Conjugate pair in
frequency domain

Periodic Noise Reduction



Bandpass and Band reject Filters

- Reject (pass) an isotropic frequency

$$H_{BP}(u,v) = 1 - H_{BR}(u,v)$$